Chronic Lymphocytic Leukaemia

Unified clinical guideline — v2.0 relapsed/refractory update

MHA-CLL-2026-v2.0
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Clinical owner: Dr Muhammad Mohsin, Consultant Haematologist

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Clinical decision-support only. Not for direct patient use. Use current SmPCs, NICE guidance, NHS commissioning rules, local policy and patient-specific clinical judgement. England-specific NICE/NHS access must not be described as a single UK-wide entitlement.

Release status

Publication authorised 27 July 2026 after independent review, pharmacy verification and clinical-owner approval of the reviewed clinical content. Production artefact hashes are controlled separately in the release record. This is educational clinical decision support, not a prescribing protocol.

Quick decision summary

- Treat only when iwCLL active-disease criteria are met — see Indications for Treatment. CLL meeting Rai/Binet criteria alone without active-disease features does not require treatment.
- TP53 abnormal (del(17p) and/or TP53 mutation) — targeted therapy, not chemoimmunotherapy. BTKi (acalabrutinib or zanubrutinib) is preferred per BSH 2025; venetoclax-based regimens are also acceptable.
- No TP53 aberration, first-line — continuous BTKi (acalabrutinib or zanubrutinib, preferred per BSH 2025) or fixed-duration venetoclax-based (venetoclax + obinutuzumab). Chemoimmunotherapy is not routine first-line in UK practice.
- At relapse or progression — confirm that iwCLL treatment criteria are met; document previous regimens, response, duration and reason for stopping; repeat del(17p) FISH and TP53 sequencing; assess for Richter transformation when clinically suspected.
- Sequence by previous exposure and reason for stopping — distinguish intolerance from progression and relapse off fixed-duration treatment from progression during active treatment. Selected venetoclax retreatment applies only after a previous response and later relapse off treatment.
- After a BTKi — pirtobrutinib is NICE-recommended under TA1173 only when covalent-BTKi retreatment, including after a fixed-duration regimen, is not clinically appropriate. Double exposure is not synonymous with double refractoriness.
- Always MDT. Escalate to consultant haematology for any treatment decision. Local trust policies and current NICE commissioning status take precedence.

Scope and Purpose

This guideline covers the diagnosis, risk stratification, and management of chronic lymphocytic leukaemia (CLL) in adults aged ≥18 years. It integrates recommendations from ELN/ESMO, BSH and ASH/NCCN with explicit separation of international evidence from NICE recommendations and NHS access in England. Access in devolved nations requires separate verification.

The guideline applies to primary and relapsed/refractory CLL in outpatient haematology, day unit, and inpatient settings. Small lymphocytic lymphoma (SLL) shares the same biological entity and management principles. Prolymphocytic leukaemia and Richter's transformation are addressed under special populations.

England — NICE-recommended CLL regimens

The following recommendations apply in England within their exact NICE populations and current NHS commissioning criteria. Marketing authorisation, NICE recommendation and operational access must be checked separately. Devolved-nation access requires separate verification:

- TA429 — Ibrutinib for previously treated CLL and restricted untreated TP53-abnormal disease (de-prioritised for new starts by BSH 2025).
- TA689 — Acalabrutinib monotherapy for previously treated CLL and specified untreated populations.
- TA931 — Zanubrutinib monotherapy for R/R CLL and specified untreated populations.
- TA561 — Venetoclax + rituximab after at least one previous therapy.
- TA796 — Venetoclax monotherapy only within its restricted molecular and prior-treatment criteria.
- TA1173 — Pirtobrutinib after a BTKi when retreatment with a covalent BTKi, including after a fixed-duration regimen, is not clinically appropriate.
- Untreated only: TA1119 — venetoclax + obinutuzumab; TA891 — ibrutinib + venetoclax.

England-specific summary. Apply the exact NICE recommendation and current NHS commissioning/Blueteq criteria; do not infer access from marketing authorisation alone.

Clinical Overview and Epidemiology

Clinical Vignette
A 68-year-old man is referred by his GP with a persistent lymphocytosis of 18×10⁹/L noted incidentally on a routine blood test. He is asymptomatic with no lymphadenopathy, weight loss, or night sweats. His blood film shows small mature-appearing lymphocytes with smear cells. He is concerned about leukaemia and asks whether he needs chemotherapy.

CLL is a chronic, treatable B-cell malignancy with a highly variable disease course. Long remissions are common with modern targeted agents, and some patients achieve undetectable MRD. However, disease control rather than cure is the current aim of standard therapy outside clinical trials. CLL should not be described as simply "incurable" — the nuanced truth is that some patients never require treatment, many achieve excellent and prolonged responses, and allogeneic transplantation or experimental approaches offer potential cure for a selected minority. For many patients, CLL behaves as a chronic disease compatible with prolonged survival, often with extended treatment-free periods or durable responses to therapy. However, outcomes are heterogeneous — a minority experience aggressive disease, Richter's transformation, or serious treatment-related complications.

CLL is the most common leukaemia in the Western world, with an incidence of approximately 5 per 100,000 per year in the UK. It predominantly affects older adults — median age at diagnosis is 70 years — with a 2:1 male predominance. CLL and SLL represent the same disease: CLL is defined by $\geq 5\times10^9$/L circulating B lymphocytes with CLL immunophenotype; SLL by predominantly nodal disease with $<5\times10^9$/L circulating CLL cells.

The disease course is highly variable — some patients live for decades without treatment, while others require early intervention and have rapidly progressive disease. This biological heterogeneity is now well-characterised by molecular markers (IGHV, TP53, del(17p)) and determines both prognosis and treatment selection.

Staging Systems

Stage	Binet	Rai	Median Survival (historical — pre-targeted therapy era)
Early	A — <3 areas, Hb ≥10, plt ≥100	0 — lymphocytosis only	>10 years
Intermediate	B — ≥3 areas, Hb ≥10, plt ≥100	I/II — adenopathy/splenomegaly	5–7 years
Advanced	C — Hb <10 or plt <100	III/IV — anaemia/thrombocytopenia	2–4 years

Staging caveat (important): Rai and Binet staging remain clinically useful for bedside communication and trial eligibility, but they are insufficient on their own for treatment stratification in the targeted-therapy era. Modern treatment selection integrates Binet/Rai with TP53 status, IGHV mutation status, fitness assessment, and patient preference. A Binet A patient with del(17p)/TP53 mutation who develops progressive cytopenia requires the same targeted approach as advanced-stage disease.

Pattern recognition: Staging alone no longer drives treatment decisions in the targeted therapy era — molecular profile and fitness assessment are equally important. Binet/Rai staging remains relevant for trial eligibility and audit purposes.

Guideline Basis (Methodology)

This guideline integrates three international frameworks using GRADE (Grading of Recommendations Assessment, Development and Evaluation) methodology. Each recommendation is tagged with its source society, evidence quality, and recommendation strength.

Source Societies

- ELN — via ESMO 2024 CLL guidance; European consensus
- BSH 2025 — UK national guideline; primary driver for UK practice
- ASH/NCCN 2024 — American guidance; educational context for UK
- iwCLL 2018 — Diagnostic and response criteria

Evidence Grading

- High — RCTs, meta-analyses with consistent results
- Moderate — RCTs with limitations or observational with strong effect
- Low — Observational or indirect evidence
- Very Low — Case series, expert opinion

Diagnosis and Initial Work-up

CLL is diagnosed by peripheral blood flow cytometry demonstrating a clonal B-cell population with characteristic immunophenotype. Bone marrow biopsy is not required for diagnosis but may be performed to assess marrow infiltration or to exclude concurrent haematological disorders.

Diagnostic Criteria (iwCLL 2018)

Diagnostic Minimum Requirements

- Peripheral blood B-lymphocyte count ≥5×10⁹/L persisting for ≥3 months
- Clonality confirmed by flow cytometry — co-expression of CD5, CD19, CD23 with weak surface immunoglobulin and either κ or λ light-chain restriction
- Characteristic CLL morphology on blood film — small mature lymphocytes with smear/smudge cells
- Matutes score ≥4 (CD5+, CD23+, CD79b weak/negative, FMC7 negative, sIg weak)

Distinguishing CLL from Monoclonal B-cell Lymphocytosis (MBL)

A frequently missed distinction in practice: if the clonal B-cell count is below 5×10⁹/L with no lymphadenopathy, organomegaly, or cytopenias attributable to the clone, the diagnosis is Monoclonal B-cell Lymphocytosis (MBL), not CLL — even if the immunophenotype is identical. MBL progresses to CLL requiring treatment at approximately 1–2% per year. Clinical low-count MBL (clonal B cells <0.5×10⁹/L) carries negligible transformation risk and warrants no haematology follow-up. High-count MBL (0.5–5×10⁹/L) warrants 6–12 monthly monitoring. The distinction matters: MBL patients should not receive CLL-directed treatment.

Pattern recognition: Same immunophenotype, different threshold. MBL = clonal B cells <5×10⁹/L with no disease features. CLL = ≥5×10⁹/L or any count with lymphadenopathy/organomegaly/cytopenia attributable to the clone (i.e., tissue disease = SLL regardless of count).

Mandatory Initial Investigations

Investigation	Purpose	Key Finding in CLL
FBC + blood film	Confirm lymphocytosis; exclude pseudo-lymphocytosis; smear cells	Lymphocytosis ≥5×10⁹/L; smudge/smear cells characteristic of CLL but not specific enough to establish diagnosis without immunophenotyping

Investigation	Purpose	Key Finding in CLL
Flow cytometry	CLL immunophenotype (Matutes score)	CD5+/CD19+/CD23+; weak sIg; FMC7 negative
Immunoglobulins	Baseline; hypogammaglobulinaemia risk	Reduced IgG/IgA/IgM — increased infection risk
DAT (Direct Antiglobulin Test)	Baseline AIHA screen	Positive in ~7% at diagnosis
LDH, β2-microglobulin	Prognostic markers	Elevated correlates with advanced disease
Virology (HIV, HBV, HCV, CMV, EBV)	Mandatory before immunosuppression; reactivation risk	Screen all; HBV core Ab especially important before anti-CD20

Pre-Treatment Imaging

CT chest/abdomen/pelvis is not mandatory at diagnosis in asymptomatic CLL but is required before initiating treatment. It is used to assess lymphadenopathy burden, identify sites of bulky disease relevant to TLS risk stratification, evaluate splenomegaly, and provide a baseline for response assessment. In patients managed with watch and wait, repeat imaging is not routinely required unless clinical change suggests disease progression or Richter's transformation.

Common pitfall: A positive DAT at diagnosis does not mean AIHA is present — check reticulocyte count, LDH, and indirect bilirubin to confirm haemolysis. A positive DAT with a normal Hb and reticulocyte count requires no immediate action beyond monitoring.

Risk Stratification (Prognostic Work-up & Molecular Markers)

Molecular profiling is mandatory before treatment but is not required at diagnosis in patients managed with watch and wait. Repeat del(17p) FISH and TP53 mutation analysis before each treatment line because TP53 status may evolve. IGHV mutation status is stable and should be established once before treatment if no valid result is already available; it does not require routine repeat testing at relapse. Results directly determine treatment selection, eligibility for targeted agents, and the risk-benefit balance.

Mandatory Prognostic Panel — Required Before Treatment

- FISH panel — del(17p), del(11q), trisomy 12, del(13q): hierarchical prognostic impact; del(17p) worst
- TP53 mutation sequencing — TP53 mutation = same prognostic and treatment implications as del(17p); must always test alongside FISH (FISH alone misses ~40% of TP53 aberrancy)
- IGHV mutation status — establish once before treatment if no valid result is available; do not routinely repeat at relapse. Unmutated IGHV (<2% divergence from germline) is associated with shorter remissions across several approaches and informs treatment selection, not treatment timing.
- Karyotype / FISH — complex karyotype (≥3 abnormalities) = inferior outcomes on all therapies including BTKi and venetoclax combinations; trial enrolment preferred
- CD38, ZAP-70 — surrogate markers; less actionable in the targeted therapy era but useful for overall risk stratification

Treatment-predictive, not merely prognostic. TP53 status and IGHV mutational status are routinely described as "prognostic markers" but this framing is incomplete. Their primary clinical importance is predictive — they forecast which patients are unlikely to benefit from specific treatment approaches and therefore materially change the therapeutic choice:

• TP53 disruption (del(17p) and/or TP53 mutation): predicts poor or no response to DNA-damaging cytotoxic approaches; mandates the use of targeted therapy (BTKi or venetoclax-based), with agent selection guided by fitness, comorbidity, and patient factors.

• IGHV unmutated: predicts a more aggressive course and shorter remissions across most therapies; does not itself trigger treatment, but informs duration and depth of response expected with each approach.

• IGHV mutated: predicts deeper and more durable remissions — particularly with fixed-duration venetoclax combinations — and is therefore relevant to treatment selection, not timing.

Neither marker alone constitutes an indication to start treatment in an otherwise asymptomatic patient.

Prognostic Impact Summary

Marker	Favourable	Unfavourable	Treatment Impact
TP53/del(17p)	Absent	Present	Avoid chemotherapy — must use BTKi or venetoclax-based; avoid ibrutinib+venetoclax TA891 data limited here
del(11q)	Absent	Present	Responds well to BTKi therapy; venetoclax combinations also effective
trisomy 12	—	Intermediate	Good response to BTKi; associated with NOTCH1 mutation
del(13q) only	Present (sole abnormality)	—	Favourable — respond to all therapies
IGHV mutated	Mutated (≥2% divergence)	Unmutated (<2%)	Mutated IGHV: deeper and more durable responses; consider fixed-duration venetoclax strategy
Complex karyotype	Absent	≥3 abnormalities	Inferior PFS on all therapies; prioritise novel agents; trial enrolment

Mnemonic — TP53 testing: "17 and TP — always together". del(17p) alone misses ~40% of TP53-aberrant cases. FISH and sequencing must always be performed together. Both del(17p) and TP53 mutation confer the same clinical management implications.

Indications for Treatment (iwCLL Active Disease Criteria)

The majority of newly diagnosed CLL patients do not require immediate treatment. Watch-and-wait is appropriate for asymptomatic patients regardless of lymphocyte count. Treatment is initiated only when active disease criteria are met.

iwCLL 2018 Active Disease Criteria — Treat When ANY Present

- Progressive marrow failure: Worsening anaemia (Hb <10 g/dL) or thrombocytopenia (plt $<100\times10^9$/L) due to CLL infiltration
- Massive or progressive splenomegaly: >6 cm below left costal margin, or symptomatic
- Massive or progressive lymphadenopathy: Longest diameter >10 cm, or symptomatic compression
- Progressive lymphocytosis: >50% increase over 2 months, or lymphocyte doubling time <6 months (if $>30\times10^9$/L)
- Autoimmune cytopenias: AIHA or ITP poorly responding to corticosteroids
- Constitutional symptoms: Weight loss >10% in 6 months; extreme fatigue; fevers >38°C for ≥2 weeks without infection; night sweats >1 month without infection

Watch and Wait — Appropriate When

- Asymptomatic Binet A or B
- Stable lymphocyte count (DT >12 months)
- No constitutional symptoms
- No progressive cytopenia
- Reassure: early treatment does NOT improve survival

Do NOT Treat — Avoid Pitfall

- Lymphocyte count alone (>30, $>100\times10^9$/L) without other criteria
- Positive prognostic markers alone (unmutated IGHV, del(17p) in asymptomatic patient)
- Patient or GP anxiety — education and reassurance preferred
- Hypogammaglobulinaemia — treat with Ig replacement, not CLL therapy

IGHV mutation status does not determine when to start treatment — it informs prognosis and treatment selection once the decision to treat has been made on clinical grounds. An IGHV-unmutated patient without active disease criteria should remain on watch and wait, not be treated early to prevent progression.

Fitness Assessment Before Treatment

All patients meeting treatment criteria should undergo formal fitness assessment before starting therapy. This determines treatment intensity and agent selection:

Assessment Tool	Purpose	Threshold
CIRS (Cumulative Illness Rating Scale)	Comorbidity burden; determines fitness for intensive therapy	CIRS ≤6 with no single organ score >2 = fit for intensive regimens
eGFR / creatinine clearance	Renal function — affects venetoclax TLS risk; BTKi preferred if eGFR <30	eGFR ≥70 mL/min generally supports use of all standard agents; eGFR <30 — BTKi preferred; venetoclax requires careful TLS risk assessment and monitoring
ECOG performance status	Functional capacity	ECOG 0–1 = fit; ECOG ≥2 consider reduced intensity or BTKi monotherapy
Cardiac assessment	Prior to BTKi — AF risk, anticoagulation status	Prior AF, anticoagulation = prefer acalabrutinib or zanubrutinib over ibrutinib

First-line Treatment

First-line treatment selection is guided by: (1) presence of del(17p) or TP53 mutation, (2) patient fitness (CIRS/eGFR/ECOG), and (3) patient preference (fixed-duration vs. continuous therapy). BSH 2025 endorses zanubrutinib and acalabrutinib as the preferred BTK inhibitors for new patients, with venetoclax-based fixed-duration combinations as the main alternative. Ibrutinib remains NICE-approved but is de-prioritised due to its inferior cardiac safety profile.

7a. TP53-Aberrant CLL — del(17p) and/or TP53 Mutation

TP53-Aberrant CLL — Targeted Agents Only (BSH 2025, GRADE 1A)

In del(17p) and/or TP53-mutant CLL, only targeted agents (BTK inhibitors and venetoclax-based combinations) achieve meaningful responses. DNA-damaging cytotoxic regimens are not appropriate in this setting and should not be used.

Zanubrutinib TA931

Dose: 160 mg twice daily (or 320 mg once daily) continuously. NICE TA931 applies only within its specified untreated populations; it also includes R/R CLL. BSH 2025 preferred BTKi for new patients alongside acalabrutinib (GRADE 1A). Trial basis: SEQUOIA — Arm B: zanubrutinib vs bendamustine-rituximab (historical control arm), HR 0.29 ($p<0.0001$); Arm C (del(17p)): 5-year PFS 72.2%. ALPINE trial (R/R): zanubrutinib superior to ibrutinib (PFS HR 0.65; $p=0.002$); in del(17p)/TP53: HR 0.53 (95% CI 0.31–0.88; primary publication NEJM 2023). Extended follow-up (39 months): HR 0.52. Lowest AF rate of all BTKis (~2%). No mandatory dose reduction for renal impairment.

Acalabrutinib (monotherapy) TA689

Dose: 100 mg twice daily continuously. NICE TA689 applies only within its specified untreated populations; it also includes previously treated CLL. Trial basis: ELEVATE-TN — 4-year PFS superior to Clb-Obi. Preferred over ibrutinib (BSH 2025 GRADE 1A) — significantly lower rates of atrial fibrillation (6% vs 11%), hypertension, and major bleeding. Suitable for most fitness levels including those on anticoagulation.

Venetoclax + Obinutuzumab (fixed 12 cycles) TA1119

Schedule: Obinutuzumab cycle 1 day 1, 100 mg; day 2, 900 mg; days 8 and 15, 1,000 mg; then 1,000 mg on day 1 of cycles 2–6. Start the five-week venetoclax ramp-up (20→50→100→200→400 mg) on cycle 1 day 22 and continue venetoclax through the end of cycle 12, subject to the current SmPC and local TLS protocol. NICE TA1119 (updates TA663). Trial basis: CLL14 (unfit/comorbid patients) — 6-year PFS median 76.2 months vs 36.4 months; uMRD at EOT 78%. GAIA/CLL13 (fit patients without TP53 aberrations) — uMRD 86.5% at month 15, 3-year PFS 90.5% for Ven-Obi. Both support fixed-duration venetoclax+obinutuzumab. Deepest remissions in IGHV-mutated disease. TLS risk: Stratify by lymph node size and WBC; inpatient ramp-up if high risk.

Ibrutinib + Venetoclax (fixed-duration) TA891

Schedule: Ibrutinib 420 mg once daily for three lead-in cycles, followed by ibrutinib 420 mg once daily plus venetoclax, ramped to 400 mg once daily, for 12 combination cycles; total treatment duration is 15 cycles. NICE TA891 was published in May 2023. Trial basis: CAPTIVATE and GLOW studies. High uMRD rates; fixed-duration advantage. Note: Data in del(17p) patients are limited — BSH guidance suggests venetoclax+obinutuzumab or BTKi monotherapy may be preferred in TP53-aberrant disease.

Ibrutinib TA429 (de-prioritised — see note)

Dose: 420 mg once daily continuously. NICE TA429 remains current. BSH 2025: Ibrutinib is de-prioritised for new patients in favour of zanubrutinib and acalabrutinib (GRADE 1A) due to higher rates of AF (~11%), hypertension (~55%), and bleeding. Appropriate to continue in patients already established on ibrutinib who are tolerating it well. Check Blueteq eligibility criteria for new starts — zanubrutinib/acalabrutinib are preferred for Blueteq approval.

7b. CLL Without TP53 Aberrancy — Treatment Selection

UK Practice — First-Line Regimen Selection (BSH 2025)

BSH 2025 (GRADE 1A): Zanubrutinib (TA931) and acalabrutinib (TA689) are the preferred BTK inhibitors for new patients, preferred over ibrutinib due to superior safety profiles. Venetoclax-based fixed-duration combinations are the main alternative. Selection is guided by:

- BTKi preference: Zanubrutinib 160 mg BD or acalabrutinib 100 mg BD — preferred for most new patients (lower AF, hypertension, bleeding vs ibrutinib)
- Fixed-duration option: Venetoclax + obinutuzumab (TA1119, 12 cycles) — preferred for patients wanting treatment-free remission; particularly effective in IGHV-mutated disease
- Cardiac risk / anticoagulation: Zanubrutinib or acalabrutinib preferred (lowest AF rates among BTKis); venetoclax-based also appropriate
- IGHV-mutated + fit: Consider venetoclax + obinutuzumab (TA1119) for deepest uMRD and treatment-free remission
- Renal impairment (eGFR <30): BTKi preferred — venetoclax requires careful TLS assessment and dose modification
- Ibrutinib (TA429/TA891): Remains NICE-approved but de-prioritised by BSH 2025 for new patients. Still appropriate if already established and tolerating well, or per Blueteq eligibility criteria.

Relapsed / Refractory CLL

Clinical Vignette
A 74-year-old woman with CLL treated with ibrutinib 420 mg OD for 3 years develops new cervical lymphadenopathy and a rising lymphocyte count of 45×10⁹/L. She has mild AF and is on apixaban. Repeat FISH shows del(17p) not previously detected. What is the next step?

At each relapse, confirm that iwCLL treatment criteria are met and distinguish progression from intolerance. Record every previous class, regimen, depth and duration of response, reason for stopping and treatment-free interval. Repeat del(17p) FISH and TP53 sequencing before the next treatment line. Assess rapidly progressive, asymmetric or highly symptomatic disease for Richter transformation with PET-CT-directed biopsy.

Define the treatment state before selecting therapy

- BTKi or BCL2-inhibitor naive: the class has not been used.
- Previously exposed, not refractory: treatment was completed or stopped without progression and sensitivity may remain.
- Intolerant: treatment stopped for toxicity without progression; a better-tolerated agent within the class may remain appropriate.
- Refractory: progression occurred during active treatment or there was no meaningful response.
- Double exposed: both a covalent BTKi and venetoclax have been used.
- Double refractory: disease has progressed during both classes. This is a distinct high-risk state and is not synonymous with double exposure.

NICE recommendations and NHS England access at 26 July 2026

Venetoclax + Rituximab (fixed-duration, R/R) TA561

Access: NICE TA561 after at least one previous therapy. Schedule: complete the licensed five-week venetoclax ramp-up to 400 mg once daily before rituximab; rituximab 375 mg/m² on cycle 1 day 1 then 500 mg/m² on day 1 of cycles 2–6; continue venetoclax for 24 months from cycle 1 day 1 of rituximab. Apply current SmPC tumour-lysis risk assessment, prophylaxis, monitoring and interaction requirements. Evidence: MURANO supports durable fixed-duration treatment. Retreatment evidence is limited to selected previous responders who relapsed after stopping therapy; it does not support routine retreatment after primary resistance or progression during active venetoclax.

Venetoclax monotherapy (post-BTKi R/R) TA796

Schedule: licensed five-week ramp-up to 400 mg once daily; continue until progression or unacceptable toxicity. Apply current SmPC tumour-lysis risk assessment, prophylaxis, monitoring and interaction requirements. TA796 recommends venetoclax monotherapy for adults with CLL: (a) with del(17p) or TP53 mutation when a B-cell receptor pathway inhibitor is unsuitable, or whose disease has progressed after a B-cell receptor pathway inhibitor; or (b) without del(17p) or TP53 mutation whose disease has progressed after both chemoimmunotherapy and a B-cell receptor pathway inhibitor. Do not reduce this to "post-BTKi", "post-CIT" or "restricted criteria".

Zanubrutinib or Acalabrutinib (BTKi-naive R/R) TA931 TA689

For BTKi-naive previously treated CLL, zanubrutinib 160 mg twice daily or 320 mg once daily (TA931) and acalabrutinib 100 mg twice daily (TA689) are NICE-recommended options, subject to current SmPC and Blueteq criteria. ALPINE established zanubrutinib's superiority to ibrutinib in R/R CLL but did not establish another covalent BTKi as treatment for disease progressing through covalent BTK inhibition. A within-class switch may be reasonable for intolerance; do not conflate this with treatment of biological resistance.

Pirtobrutinib (non-covalent BTKi) TA1173

Access: TA1173 recommends pirtobrutinib for adults with relapsed or refractory CLL who have had a BTK inhibitor, only if retreatment with a covalent BTK inhibitor, including retreatment after a fixed-duration regimen, is not clinically appropriate. At 26 July 2026, NHS England access remained available through interim Blueteq funding during implementation of TA1173; routine-budget implementation was due by 29 September 2026. The broader MHRA licence does not widen TA1173 or NHS funding eligibility. Schedule: 200 mg once daily continuously, with dose modification according to the current SmPC. Evidence: BRUIN phase I/II reported independent-review ORR 73.3% after covalent BTKi. Randomised BRUIN CLL-321 showed median PFS 14.0 versus 8.7 months against idelalisib–rituximab or bendamustine–rituximab (HR 0.54); no overall-survival advantage was demonstrated at the reported analysis.

Sequencing after prior targeted therapy

Prior-treatment state	Preferred decision route	Boundary
Covalent-BTKi intolerance without progression	Consider a better-tolerated covalent BTKi or a different class according to comorbidity, preference and NICE eligibility.	A within-class switch for intolerance is not evidence of efficacy after resistance.
Covalent-BTKi progression; venetoclax naive	Use a venetoclax-based NICE pathway when eligible; pirtobrutinib is an alternative under TA1173 when its restriction is met.	Do not continue a covalent BTKi as sole definitive therapy for resistant disease.
Previous fixed-duration venetoclax response; relapse off treatment	Consider a BTKi or selected venetoclax-based retreatment after reviewing prior response, treatment-free interval and current sensitivity.	Retreatment evidence does not cover primary or on-treatment resistance.
Venetoclax progression; BTKi naive	Use a NICE-funded covalent BTKi when eligible.	No randomised trial establishes a universal sequencing hierarchy.
Double exposed, not double refractory	Individualise according to retained class sensitivity, duration of prior response and TA eligibility.	Do not automatically label as double refractory.
Double refractory	Early tertiary review and clinical-trial referral; use pirtobrutinib when TA1173 criteria are met; consider cellular therapy or allogeneic transplantation only in highly selected patients through specialist pathways.	No proven comparative hierarchy. No demonstrated CLL marketing authorisation, NICE recommendation or routine NHS England commissioning for CAR-T at the cut-off; devolved-nation access was not established by this audit.

Emerging options are not routine NHS treatment

- Pirtobrutinib + venetoclax + rituximab: BRUIN CLL-322 reported improved PFS over venetoclax–rituximab, but prior BCL2 inhibitor or non-covalent BTKi exposure was excluded. NICE ID6566/GID-TA11748 remained awaiting development at the evidence cut-off. Do not imply routine access.
- Lisocabtagene maraleucel: TRANSCEND CLL 004 provides single-arm efficacy evidence after BTKi progression and venetoclax failure. There was no demonstrated CLL marketing authorisation, NICE recommendation or routine NHS England commissioning at the cut-off; devolved-nation access was not established by this audit. NICE ID6174/GID-TA11001 was discontinued in February 2026.
- CD20×CD3 bispecific antibodies: no qualifying peer-reviewed prospective therapeutic trial in untransformed CLL was identified through 26 July 2026. Do not extrapolate Richter-transformation or other-lymphoma evidence.

BTK Inhibitor Resistance — Key Molecular Mechanisms

- BTK C481S mutation — most common; prevents covalent binding of ibrutinib/acalabrutinib/zanubrutinib; pirtobrutinib active
- PLCG2 mutations — downstream signalling may confer resistance; do not assume equivalent sensitivity across all non-covalent BTK inhibitors.
- Clonal evolution — repeat del(17p) FISH and TP53 sequencing before each new treatment line.
- Possible Richter transformation — suspect with sudden deterioration, disproportionate LDH rise or rapidly enlarging asymmetric disease; obtain PET-CT-directed biopsy.

Response & Monitoring (Response Assessment and MRD)

Response	Definition (iwCLL 2018)
Complete Response (CR)	No lymphadenopathy >1.5 cm, no splenomegaly, no B-symptoms, ALC $<4\times10^9$/L, plt ≥100, Hb ≥11 (untreated by growth factors), BM normocellular without CLL nodules
Partial Response (PR)	≥50% reduction in lymphadenopathy/organomegaly + improvement in cytopenias
Stable Disease (SD)	<50% reduction in measurable disease
Progressive Disease (PD)	≥50% increase in measurable disease or new disease manifestation

Note on MRD: MRD assessment is not a formal iwCLL response category (the 2018 iwCLL criteria define CR, PR, SD, PD based on clinical and haematological parameters). MRD status — assessed by flow cytometry or next-generation sequencing — is evaluated as an additional layer to guide treatment duration in fixed-duration regimens.

MRD Assessment

Undetectable MRD (uMRD4), defined as fewer than 1 CLL cell per 10,000 leucocytes ($<10^{-4}$) by flow cytometry or next-generation sequencing, is an emerging surrogate endpoint used to guide treatment duration in fixed-duration regimens. uMRD at end of treatment in CLL14 correlated strongly with prolonged treatment-free remission.

Current UK practice: MRD assessment is not yet mandated for routine clinical decision-making outside clinical trials in the UK. It is used in the context of research and to inform fixed-duration treatment stopping rules. BSH recommends MRD assessment in trial settings.

Special Populations

10a. Richter's Transformation (RT)

Richter's transformation occurs in 2–10% of CLL patients at an annual incidence of 0.5–1%. In ~90% of cases the histology is DLBCL (clonally related in ~80%); the remainder are Hodgkin lymphoma. Median OS post-RT is approximately 9 months with conventional therapy — overall prognosis is poor.

Suspect Richter's Transformation When

- Sudden deterioration with new or rapidly growing lymph node (often asymmetric)
- Markedly elevated LDH (disproportionate to lymphocyte count)
- B-symptoms emerging or worsening without intercurrent infection
- PET-CT: highly avid FDG uptake in single or asymmetric nodal group (SUVmax >5)

Action: PET-CT + biopsy of most metabolically active node. Do NOT rebiopsy the same site if initial biopsy was non-diagnostic — biopsy the most active PET site.

RT Management

- Enrol on clinical trial where possible — no standard of care
- R-CHOP if no trial available — inferior outcomes compared with de novo DLBCL
- CAR-T (CD19-directed) — ORR 60–65%; responses can be durable
- Bispecific antibodies (epcoritamab, mosunetuzumab) — promising in R/R RT; check TA status
- ASCT consolidation for chemosensitive RT in eligible patients
- Palliative intent if poor PS or major comorbidity

Boundary: These statements apply to Richter transformation and must not be extrapolated to untransformed CLL.

Autoimmune Cytopenias in CLL

- Autoimmune cytopenias (AIHA, ITP) occur in up to 20% of CLL patients during their illness course (BSH 2012: 10–20%; AIHA specifically ~7%; DAT positive without haemolysis in a further 7–14%)
- Diagnose AIHA: DAT positive + haemolysis (elevated LDH, bilirubin, reticulocyte count)
- First-line: Manage as for primary autoimmune cytopenia — corticosteroids (prednisolone 1 mg/kg, tapering over 2–4 weeks) are the usual initial treatment
- CLL-directed therapy is indicated when: the autoimmune complication is refractory to steroids, recurrent, steroid-dependent, or occurs in the context of active/progressive CLL requiring treatment in its own right
- If steroids fail or CLL therapy not yet due: rituximab 375 mg/m² ×4
- Avoid fludarabine-containing regimens in active AIHA — can worsen haemolysis
- Evans syndrome (AIHA + ITP): higher complexity — specialist input

10b. CLL in Pregnancy

CLL in Pregnancy — Management Principles

- Observation: Most CLL in pregnancy does not require CLL-directed treatment — indolent course permits deferral until postpartum in most cases
- Symptomatic hypogammaglobulinaemia: IVIG safe in pregnancy; subcutaneous Ig alternative
- If treatment required: Multidisciplinary approach (haematology + obstetrics + neonatology); consider timing (avoid 1st trimester if possible)
- Avoid BTK inhibitors: Limited safety data; teratogenicity potential in animal studies — risk/benefit discussion required
- Avoid venetoclax: No adequate human pregnancy data; embryotoxic in animal models
- If urgent treatment needed: Anti-CD20 ± corticosteroids or single-agent rituximab (neonatal B-cell depletion reported — notify neonatology)
- Neonatal monitoring: Check cord blood for thrombocytopenia; neonatal B-cell monitoring if maternal anti-CD20 received

10c. Hypogammaglobulinaemia and Infection Prophylaxis

Infection Prophylaxis in CLL

- Immunoglobulin replacement (IgRT): IVIG or SCIG when IgG <4 g/L with recurrent bacterial infections (≥2 significant infections per year) — not for IgG low without clinical infection
- Pneumocystis jirovecii prophylaxis: Co-trimoxazole 480 mg BD (3 days/week) during and for 6–12 months after purine analogue, venetoclax, or anti-CD20 therapy
- Herpes simplex/zoster prophylaxis: Aciclovir 400 mg BD (or valaciclovir 500 mg BD) during and for ≥6 months after venetoclax-based combinations or intensive anti-CD20 therapy; consult local protocol — BSH guidance does not mandate it for BTKi monotherapy but it is commonly prescribed
- CMV monitoring: Routine weekly CMV PCR is an alemtuzumab-era protocol — alemtuzumab is no longer used in CLL in the UK. Targeted therapies (BTKis, venetoclax) carry significantly lower CMV reactivation risk. CMV monitoring is not routinely required for patients on BTKi monotherapy or venetoclax. Consider CMV surveillance in: (1) venetoclax + anti-CD20 combinations in significantly immunocompromised patients with known CMV seropositivity; (2) post-allogeneic SCT (a small minority of CLL patients); (3) patients with severe combined immunodeficiency or unexplained cytopenias on treatment. Follow local haematology/virology protocols.
- Vaccinations: Annual influenza (inactivated); pneumococcal (PCV13 + PPSV23); COVID-19; hepatitis B; shingles vaccine (Shingrix — recombinant; live vaccines contraindicated)
- HBV reactivation: Screen all patients — HBV core Ab positive patients require prophylactic antiviral (entecavir or tenofovir) during anti-CD20 treatment

Audit & Governance (Audit Standards)

The following quality indicators can be used to audit CLL management practice against BSH 2025 and NICE commissioning requirements.

Audit Standard	Data Source	Target	Rationale
Flow cytometry performed and documented in all newly diagnosed CLL cases	Haematology database, clinic letters	≥98%	iwCLL diagnostic requirement
TP53 sequencing AND FISH for del(17p) performed before each line of treatment	Molecular lab records, clinic letters	≥95%	BSH 2025 — TP53 alone insufficient
IGHV mutation status documented before first treatment	Molecular lab records	≥90%	Informs treatment strategy per BSH
Chemoimmunotherapy NOT used as first-line treatment unless documented rationale	Prescribing records, MDT notes	≥95%	BSH 2025: targeted agents are standard of care
NICE TA used for first-line treatment (TA931, TA689, TA1119, or TA891)	Blueteq / pharmacy records	≥95%	Blueteq/commissioning compliance — zanubrutinib (TA931) and acalabrutinib (TA689) preferred for new starts per BSH 2025
HBV status checked before anti-CD20 therapy; prophylaxis prescribed if HBcAb+	Virology, prescribing	100%	Fatal HBV reactivation risk
iwCLL active disease criteria documented as reason for treatment initiation	Clinic letters, MDT records	≥95%	Avoids premature treatment
Previous treatment class, response, duration and reason for stopping documented before each R/R treatment decision	Clinic letters, MDT records	≥95%	Distinguishes intolerance, relapse off treatment and biological resistance
TA1173 use documents previous BTKi exposure and why covalent-BTKi retreatment is not clinically appropriate	Blueteq, pharmacy and MDT records	100%	NICE eligibility and commissioning compliance

Limitations and Update Plan (part of Audit & Governance)

- TA931 and TA689 apply only within their specified untreated populations; both also include previously treated/R/R CLL as stated in their respective recommendations. Verify current Blueteq eligibility before prescribing.

- Head-to-head comparison between acalabrutinib and venetoclax+obinutuzumab in first-line is not yet available from phase III data — selection continues to be based on indirect comparisons and patient/clinician preference.
- MRD-guided treatment duration is an active research question — not yet part of routine UK clinical practice outside trials.
- Ibrutinib+venetoclax (TA891) data in del(17p)/TP53-mutant CLL are more limited — BSH guidance favours alternative approaches in this group until further data.
- Pirtobrutinib is NICE-recommended under TA1173 after a BTKi only when retreatment with a covalent BTKi, including after a fixed-duration regimen, is not clinically appropriate. The broader MHRA licence must not be presented as broader NHS funding.
- Pirtobrutinib + venetoclax + rituximab has phase III evidence from BRUIN CLL-322, but NICE ID6566/GID-TA11748 remained awaiting development at the evidence cut-off and the regimen is not presented as routine NHS treatment.
- Venetoclax retreatment data arise from small, selected previous-responder cohorts and do not support routine retreatment after primary resistance or progression during active treatment.
- Double-exposed and double-refractory disease are clinically distinct. Available cohorts do not establish a comparative hierarchy between pirtobrutinib, retreatment, cellular therapy and allogeneic transplantation.
- Lisocabtagene maraleucel has single-arm efficacy evidence in heavily pretreated CLL, but there was no demonstrated CLL marketing authorisation, NICE recommendation or routine NHS England commissioning at the cut-off; devolved-nation access was not established by this audit. The CLL NICE appraisal ID6174/GID-TA11001 was discontinued in February 2026.
- No qualifying peer-reviewed prospective therapeutic trial of the principal CD20×CD3 bispecific antibodies in untransformed CLL was identified through 26 July 2026. Richter-transformation and other-lymphoma evidence cannot be extrapolated.
- NICE and NHS England statements principally describe England. Access in Scotland, Wales and Northern Ireland requires separate verification.
- SLL (Small Lymphocytic Lymphoma) shares the same biological entity and treatment principles as CLL; high-level recommendations apply to both. Detailed SLL-specific diagnostic nuances (particularly regarding nodal biopsy, PET-CT staging, and tissue diagnosis) are not addressed here — haematology lymphoma MDT review is recommended for SLL-dominant presentations. Waldenström macroglobulinaemia is a distinct entity not covered by this guideline.

Update schedule: Evidence and access status were reviewed to 26 July 2026. Recheck NICE, NHS England and MHRA status immediately before publication and at least annually thereafter, or earlier after material guidance, licensing or commissioning changes.

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33. NICE ID6174/GID-TA11001 — lisocabtagene maraleucel for previously treated CLL; appraisal discontinued 18 February 2026.

Version history

Version	Date	Author	Change
v1.0	Apr 2025	Dr M Mohsin	Initial controlled publication.
v1.9	Apr 2026	Dr M Mohsin	Published baseline before the substantive R/R revision.
v2.0	27 Jul 2026	Dr M Mohsin	Published TA1173 and evidence-led R/R sequencing update.

Release-control record

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Pharmacy verification	COMPLETE — verifier identity retained privately
Independent clinical/content review	PASS
Clinical-owner publication authorisation	COMPLETE — 27 JULY 2026
Production artefact hashes	CONTROLLED IN RELEASE RECORD